

Hayeon Jeong

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RESEARCH INTERESTS

Visual Generative Models, Post-training for Generative Models, Mechanistic Interpretability, AI Safety/Reliability.

EDUCATION

Yonsei University

M.S. in the School of Integrated Technology

Seoul, Korea

Mar. 2025 – Feb. 2027 (Expected)

- GPA 4.16/4.30

University of California, Berkeley

Visiting Undergraduate Student

Berkeley, CA, USA

Jan. 2023 – May 2023

Soongsil University

B.S. in Electronic and Information Engineering

Seoul, Korea

Mar. 2020 – Feb. 2025

- GPA 4.11/4.50 (Magna Cum Laude)

PUBLICATIONS

1. **Hayeon Jeong** and Jong-Seok Lee (2025). “[Dominating vs. Dominated: Concept-Level Generative Collapse in Diffusion Models](#)”. arXiv preprint; under review at *ACCV 2026*.
2. Sohee Park, **Hayeon Jeong**, and Daeseon Choi (2025). “[Infrared Thermal-Guided Adversarial Patch Defense for Robust Visible Person Detectors](#)”. In: *IEEE AVSS 2025*.

RESEARCH EXPERIENCE

UCLA

Visiting Researcher

Los Angeles, CA, USA

May 2026 – Oct. 2026 (Expected)

- Part of the [CML Lab](#), advised by [Prof. Cho-Jui Hsieh](#).
- Working on *Flip What Fails* (first author): RL post-training of text-to-image models on long real-user prompts with counterfactual pairs that flip only the requirements a model fails, preserving satisfied ones.

Yonsei University

M.S. Researcher

Seoul, Korea

Mar. 2025 – Feb. 2027 (Expected)

- Part of the [MCML Lab](#), advised by [Prof. Jong-Seok Lee](#).
- Formalized concept dominance in multi-concept diffusion generation and built DominanceBench; showed via controlled fine-tuning that low training-image diversity drives it, and found through head ablation that, unlike memorization, it cannot be removed by ablating any small set of heads (**under review, ACCV '26**).

Soongsil University

Undergraduate Research Intern

Seoul, Korea

Jun. 2023 – Aug. 2024

- Part of the [AI Safety Center](#), advised by Prof. Daeseon Choi.
- Trained YOLOv5/YOLOv8 person detectors and ran the full physical-world experiment (printed patches, 20 participants, paired RGB–thermal capture) for a thermal-guided adversarial-patch defense (**AVSS '25**).
- Built an on-device privacy assistant that summarizes consent forms with LLaMA-3 8B Instruct; key-phrase marking raised precision from 77.3% to 98.8% (**Best Paper Award, KIISC Chungcheong Conference**).

SELECTED PROJECTS

- **Border Conditions in Adversarial Patch Attacks.** Showed on YOLOv5 that patch border color, thickness, and interpolation shift hiding-attack success rates by up to 9.8 points (4.5 on average); gray 4–8 px borders help most (2nd Place, Undergraduate Thesis Competition, Soongsil University). ([project page](#))
- **Phonetic-Aware Encoder Tuning for L2 Korean Speech Recognition.** Automatic speech recognition (ASR) framework robust to phonetic disruption in non-native Korean speech via a LoRA-adapted Whisper encoder with auxiliary connectionist temporal classification (CTC) supervision. ([project page](#))
- **Lightweight DenseNet.** EMNIST classifier with a DenseNet variant using $10\times$ fewer parameters than ResNet-50 and half the training time, and $2\times$ faster than DenseNet-121. ([project page](#))

HONORS & AWARDS

- **Best Paper Award (KISA President's Award),** KIISC Chungcheong Conference, 2024.
- **Professor's Scholarship** (50% tuition, merit-based), School of Electronic and Information Engineering, Soongsil University, 2024.
- **Truth Award – Professor's Scholarship** (25% tuition, merit-based), Soongsil University, 2024.
- Academic Excellence Scholarship (25% tuition, merit-based), Soongsil University, 2021.

ACADEMIC SERVICE & TEACHING

- Reviewer, *IEEE ICASSP 2026*.
- Teaching Assistant, *Advanced Engineering Mathematics*, Yonsei University, Spring 2025.

OTHER EXPERIENCE

- Trainee, SK Hynix Semiconductor Curriculum 2.0, SUNI $\times$ SK hynix (Sep. 2023 – Dec. 2023); completed coursework on the semiconductor market, memory devices (DRAM, NAND), CIS, and industry trends.
- Data Annotator, AI Training Data Construction Project, Univa (Jul. 2022 – Sep. 2022); annotated mathematical-equation images to build supervised training data for an OCR recognition model.
- Member, Baird Volunteer Corps, Soongsil University (Oct. 2020 – Feb. 2022); contributed to community safety through a CCTV installation project with the local police station and youth education programs.
- Mentor, KT Mentoring Program (Oct. 2020 – Mar. 2022); developed a 6-month Math/English curriculum for middle school students.

SKILLS

- **Languages:** Korean (native), English (professional working proficiency).
- **Programming:** Python, Java, C/C++.
- **Frameworks:** PyTorch, Hugging Face (Diffusers/Transformers).
- **Tools:** Git, LaTeX.